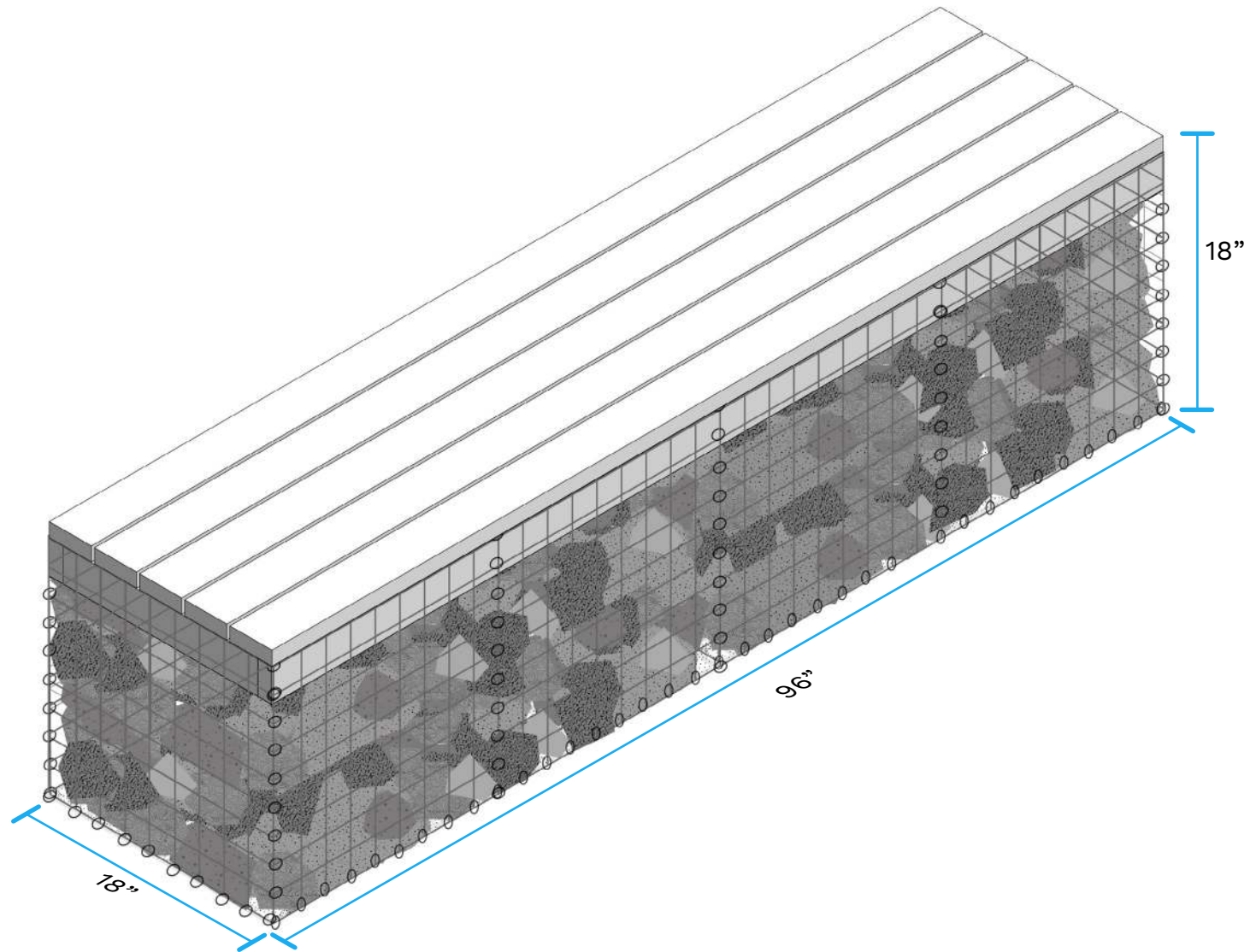




8' GABION BENCH

ASSEMBLY INSTRUCTIONS

PARK IN A TRUCK The Landscape Architecture Program + The Lab for Social and Urban Innovation Thomas Jefferson University

WORKING VERSION 1.0

BEFORE YOU BEGIN

1. This document includes recommendations for do-it-yourself (DIY) construction of a **8' gabion bench**. As a DIY project, the Owner, or the Owner's Contractor, is required to procure, fabricate, assemble, install, and maintain all project materials. These are recommendations only and any material, assembly, and construction must be tailored to specific project requirements.
2. These recommendations are based on common best practices associated with standard DIY home-improvement and/or landscape construction. These best practices may include working knowledge and experience with manual and power tools, hardware, fasteners, lumber, concrete footings, and associated techniques of assembly and construction. Prior working knowledge of these best practices is required so as to avoid harm or personal injury to self or others, or to property.
3. This document includes recommendations for the sequence and assembly of **gabion bench** components. Locate your **gabion benches** in your plan you created in the 'Dream' workbook to find the proposed location and dimension of all **gabion benches**. This plan must be tailored to the specific conditions of the project site, including property lines, existing features- such as walls, trees, etc- and any other existing site conditions that may limit or otherwise influence the installation. Use your cost estimation spreadsheet to determine the quantity needed for all your **gabion benches**.
4. Before using any power equipment (such as drills, saws, augers, or other), review all manufacturer safety requirements. While handling or operating any power equipment, follow all manufacturer safety requirements. Do not allow others who have not reviewed and/or follow manufacturer safety requirements to handle or utilize power equipment.
5. The owner is solely responsible for any work required to secure, and comply with, all necessary building and safety codes, permits, and regulatory approvals.
6. These recommendations assume that the project site has been sufficiently prepared for construction. This includes, but is not limited to: identification of utilities that may interfere or otherwise impact installation; the removal of trash, hazards, unwanted structures; or other conditions that may inhibit safe construction.
7. Before construction, review the project site conditions to identify any potential limitations to assembly and installation. This includes determining advance notification by the Owner of utility companies so as to determine the location of all above and below ground utilities.
8. All persons associated with building and construction should sign a volunteer waiver.

LEGAL NOTICE:

The recommendations included herein are based on standard practices of fabrication and construction of garden furnishings. In no event shall Thomas Jefferson University (TJU), TJU's Lab For Social Innovation, Park-in-a-Truck, and/or its faculty, staff or affiliates be liable for any damages or injury to person or property damages arising out of the use of these recommendations. Thomas Jefferson University (TJU), TJU's Lab For Social Innovation, Park-in-a-Truck, or its faculty, staff and/or affiliates make no warranties, expressed or implied, and hereby disclaims and negates all other warranties, including without limitation, implied warranties or conditions of merchantability, fitness for a particular purpose, or non-infringement of intellectual property or other violation of rights. Further, Thomas Jefferson University (TJU), TJU's Lab For Social Innovation, Park-in-a-Truck, and/or its faculty, staff or affiliates do not warrant or make any representations concerning the accuracy, likely results, or reliability of the use of these recommendations. The use of these recommendations constitutes your acknowledgment of the aforementioned.

8' GABION BENCH

ASSEMBLY INSTRUCTIONS

COMMON MATERIAL SELECTIONS \

GABION BASKET

METAL 2X2 WIRE MESH, FABRICATED WITH 8 GAUGE (3/16" DIA. WIRE) WELDED-WIRE, GALVANIZED-WITH 2" OPENINGS.

FABRICATION SUGGESTIONS:

DARBY AND SONS: WWW.DARBYWIREMESH.COM

GABION1.COM: [HTTPS://GABION1.COM/PRODUCT/18-HIGH-X-18-THICK-1/](https://GABION1.COM/PRODUCT/18-HIGH-X-18-THICK-1/)

NOTE: WHEN ORDERING PIECES, ENSURE TO STATE THE PIECES SHOULD BE CUT TO SIZE, CLOSED PIECES, GALVANIZED AFTER

SIZES: SEE STEP 01

BRACING MATERIAL

OPT 1: ROLL OF 14 OR 16 GAUGE SMOOTH, GALVANIZED WIRE

OPT 2: EXCESS PIECES OF 2X2 WIRE METAL MESH (YOU WILL HAVE EXTRA PIECES OF WIRE MESH TO CUT IN HALF IF BUILDING MORE THAN ONE GABION BASKET AT A TIME)

GABION FILL MATERIAL

STEPS 1-7 CAN BE PREFORMED OFF SITE BEFORE THEY ARE FILLED IN (STEP 9)

ALL FILL MATERIAL SHOULD BE ANGULAR IN SHAPE. MATERIALS SHOULD BE HARD AND INERT SO THAT IT WILL NOT BREAK-DOWN OVER TIME. BOTH THE FILL MATERIAL AND THE GABION PANELS WORK TOGETHER TO PROVIDE STRUCTURAL INTEGRITY. CUT LOGS SHOULD NOT BE USED FOR THIS REASON.

MATERIALS FREQUENTLY USED FOR GABION FILL INCLUDE QUARRIED ROCK, RECYCLED PAVEMENT MATERIALS (CONCRETE, ASPHALT, OTHER) OR OTHER RECYCLED MATERIALS SUCH AS BRICK OR CONCRETE MASONRY UNITS. THE SELECTION OF APPROPRIATE FILL MATERIALS IS UP TO YOU.

FILL MATERIAL SHOULD BE A MIX OF SIZES, RANGING FROM 12 INCHES MAXIMUM TO 2 INCHES MINIMUM. ALL FILL MATERIAL PIECES USED ON THE EXTERIOR FACE WILL NEED TO BE LARGER THAN 3 INCHES.. SMALLER FILL PIECES WILL BE USED TO FILL IN ANY VOIDS, AS NECESSARY.

<https://www.rockler.com/learn/choosing-the-best-finish-for-outdoor-projects>
<https://www.realmilkpaint.com/blog/tips/best-finish-outdoor-wood/>
<https://vermontnaturalcoatings.com/product-category/exterior/>

8' GABION BENCH \

TOOLS:

10" OR 12" MITER SAW (or appropriate hand saw)

IMPACT DRIVER for self drilling exterior wood screws

DRILL with 1/8" drill bit if needed for pre-drilling holes

SANDPAPER 120 GRIT

CARPENTERS SQUARE

SAFETY GOOGLES + GLOVES

HOG RING PLIERS

BOLT CUTTERS

HAMMER

MATERIALS:

(36) 2.5" SELF DRIVING EXTERIOR WOOD SCREWS

(32) CABLE STAPLES

(50) 3/4" HOG RINGS

(32) CABLE STAPLES

(36) 2.5 SELF DRIVING EXTERIOR WOOD SCREWS

COMMON LUMBER SELECTIONS \

LUMBER:

WESTERN RED CEDAR:

(8) 2" x 4" x 8'

PRO: NATURALLY WEATHER/ROT RESISTANT LUMBER WITH AESTHETICALLY PLEASING RED TONES AND PLEASANT AROMAS, THAT IS EASY TO WORK WITH. NO PRESERVATION TREATMENTS ARE NECESSARY AS THE SURFACES WILL GRADUALLY DEVELOP A LOVELY GRAY PATINA. IT IS EASILY RECYCLABLE AT THE END OF ITS PRIMARY USE.

CON: COST, BE SURE TO SPECIFY 'KNOTTY' (STK) GRADE WHICH IS MUCH MORE AFFORDABLE THAN 'CLEAR' GRADE.

DOUGLAS FIR:

PRO: MODERATELY WEATHER/ROT RESISTANT AND STRONG LUMBER. IT IS ALSO EASY TO WORK WITH AT A LOWER COST THAN CEDAR. IT IS EASILY RECYCLABLE AT THE END OF ITS PRIMARY USE.

CON: IT WILL NOT STAND THE TEST OF TIME AS WELL AS CEDAR, BUT A GREAT CHOICE ESPECIALLY IF YOU PLAN TO PAINT IT.

PRESSURE TREATED:

PRO: WEATHER/ROT RESISTANT LUMBER AT A LOW COST.

CON: LUMBER IS OFTEN OF VARIABLE POOR QUALITY AND SATURATED WITH CHEMICAL TREATMENT MAKING IT DIFFICULT TO WORK WITH LEADING TO INCONSISTENT RESULTS. NEEDS 1-4 MONTHS OF AIR DRYING BEFORE PAINTING. NOT EASILY RECYCLABLE AND OFTEN ENDS UP IN LANDFILLS AT THE END OF ITS PRIMARY USE. NOT RECOMMENDED FOR USE IN AREAS WHERE VEGETABLES ARE GROWN.

GENERAL NOTES:

1. ORDER 5-10% MORE LUMBER DEPENDING ON OVERALL QUANTITIES TO ALLOW FOR SOME VARIABLE LUMBER QUALITY AND INCORRECT CUTS.

2. EXTERIOR WOOD TREATMENT CAN PROLONG THE LIFE OF YOUR PARK ELEMENTS. HOWEVER, IT IS NOT NECESSARY TO BRING MANY YEARS OF ENJOYMENT. ANY TREATMENTS MUST BE APPLIED AND MAINTAINED PROPERLY WHICH IS A LABOR INTENSIVE PROCESS. ONE SHOULD DECIDE IF THIS IS PRACTICAL OR NOT FOR THEIR PARK. THE PARK ELEMENTS ARE LOW COST, DURABLE, AND EASY TO BUILD BY DESIGN AND IT MAY BE MORE PRACTICAL TO SIMPLY HAVE THE NEXT GENERATION REBUILD THE ELEMENTS WHEN DEEMED NECESSARY. BELOW ARE RESOURCES/ LINKS FOR CONSIDERATION IF INTERESTED IN EXTERIOR WOOD TREATMENTS.

<https://www.rockler.com/learn/choosing-the-best-finish-for-outdoor-projects>

<https://www.realmilkpaint.com/blog/tips/best-finish-outdoor-wood/>

<https://vermontnaturalcoatings.com/product-category/exterior/>

1 STEP

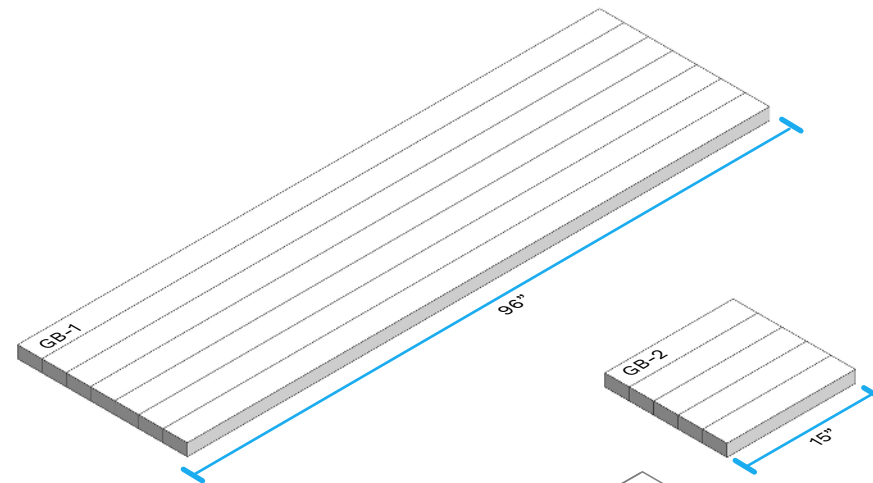
1. MARK, LABEL AND CUT LUMBER IN ORDER OF GB-1 TO GB-6 TO THE FOLLOWING LENGTHS:

NOTE: CHECK EACH PIECE OF LUMBER IMPERFECTIONS SUCH AS CRACKS, WARPING, OR UNEVEN MILLING. DISCARD THESE PORTIONS OF LUMBER.

GABION BENCH ASSEMBLY

GB-1: (7) 2 x 4 x 96"
GABION BENCH TOP

GB-2: (6) 2 x 4 x 15"
GABION BENCH TOP

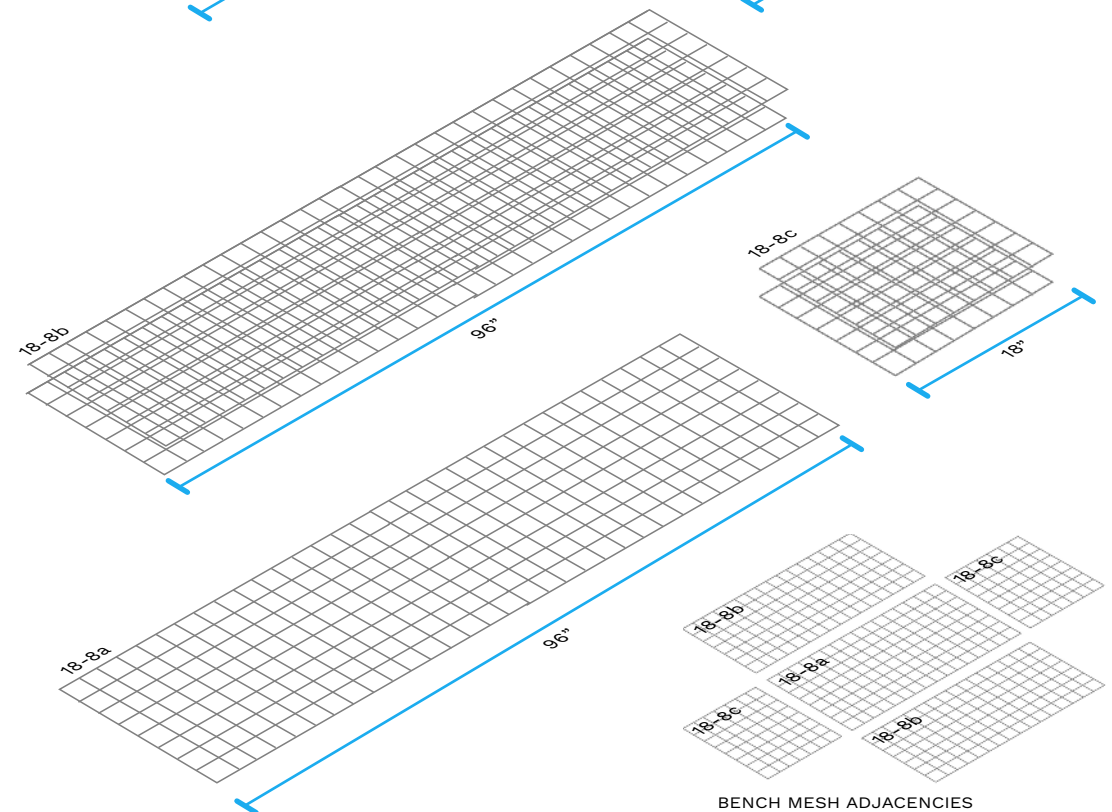


2. SORT YOUR 2x2 WIRE MESH INTO EQUAL SIZED PILES AND LABEL:

18-8a (1) 96" x 18"
GABION BOTTOM

18-8b: (2) 96" x 18"
GABION SIDE LONG

18-8c: (2) 18" x 18"
GABION SIDE SHORT



TIPS:

- LABEL EACH PIECE OF LUMBER AS YOU ARE MEASURING/CUTTING
- HAVE A PILE OF LUMBER FOR GB-1 TO GB-7
- LIGHTLY SANDING CUT EDGES TO REMOVE SPLINTERS WILL ALSO HELP WITH ASSEMBLY.
- IF BUILDING MULTIPLE GABION BENCHES, CREATING A STOP FOR YOUR MITER SAW ENSURES ALL PIECES ARE CUT TO THE SAME LENGTH.

2

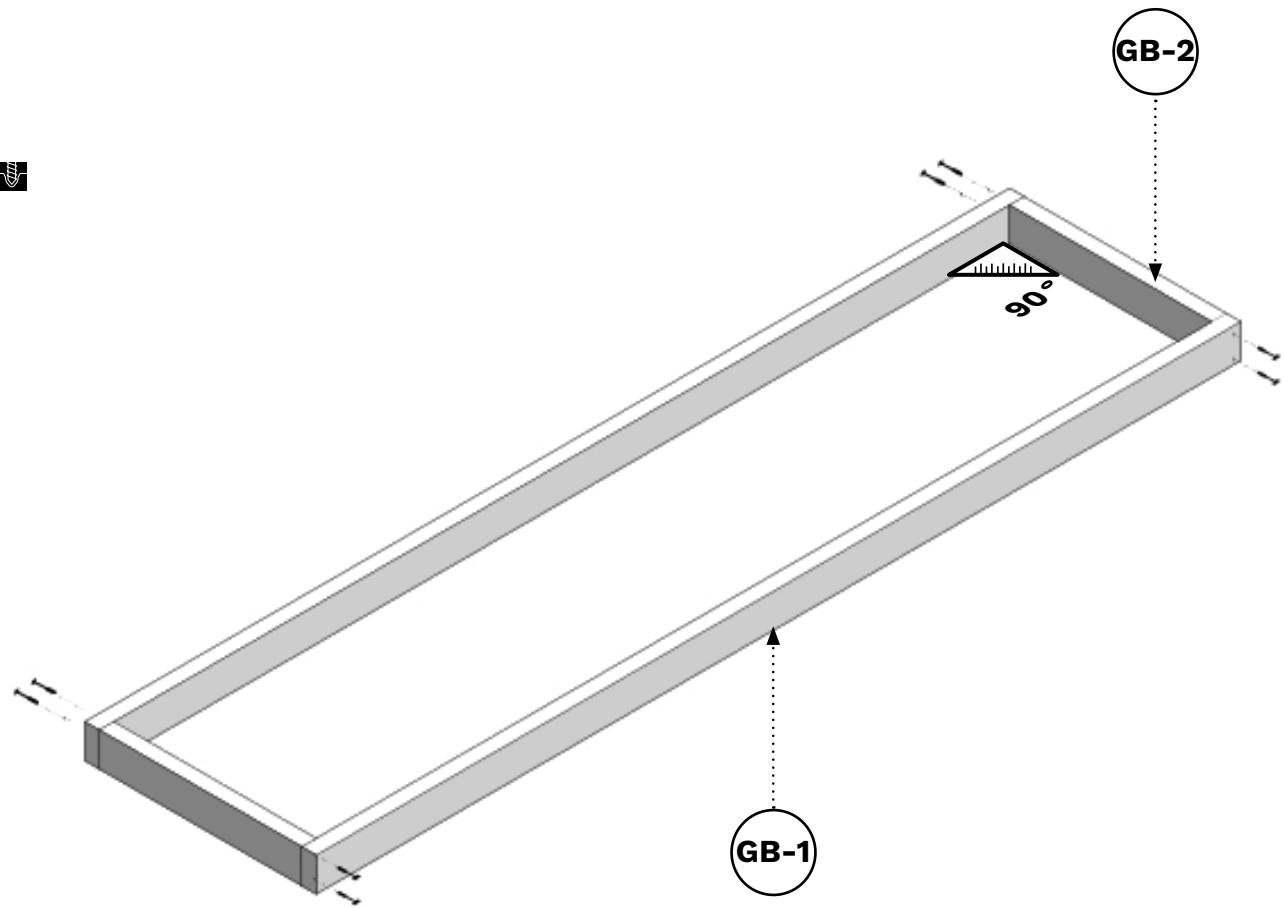
STEP

1. PLACE (2) GB-1 AND (2) GB-2 ON FLAT SURFACE. ALIGNING THE EDGES AS SHOWN IN THE IMAGE.

2. USE THE PRE-DRILLING GUIDE  AT THE END OF THE PACKET TO PRE-DRILL YOUR SCREW HOLES

3. SECURE GB-1 LUMBER INTO THE GB-2 PIECES BY USING AN IMPACT DRIVER WITH (8) 2.5" SELF DRIVING EXTERIOR WOOD SCREWS.

NOTE: UTILIZE A CARPENTERS SQUARE TO ENSURE A RIGHT ANGLE IS FORMED IN EACH CORNER. IF THE RIGHT ANGEL IS FAR OFF, CHECK THAT YOUR MEASUREMENTS OF EACH BOARD ARE CORRECT.



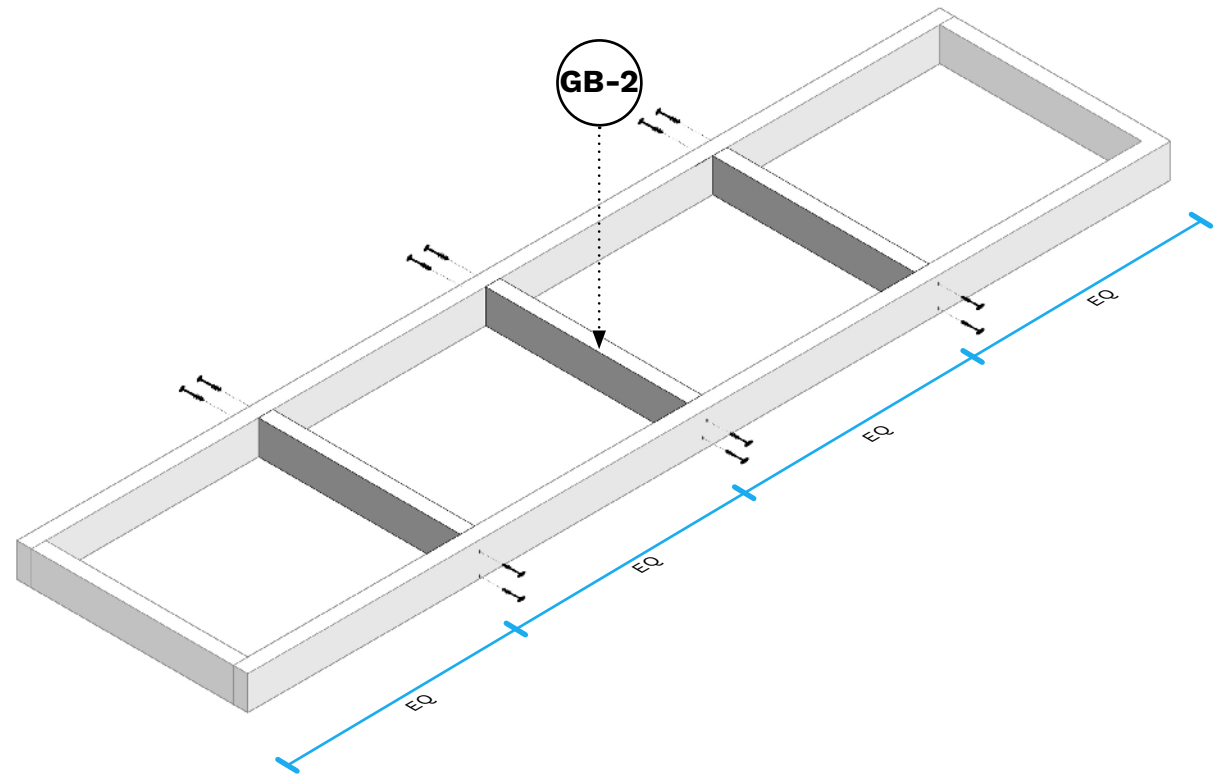
3 STEP

1. PLACE **(1)** **GB-2** PIECE IN THE MIDDLE OF THE FRAME CREATED IN STEP TWO.

2. SECURE LUMBER USING AN IMPACT DRIVER WITH **(4)** 2.5" SELF DRIVING EXTERIOR WOOD SCREWS.

3. REPEAT WITH **(2)** **ADDITIONAL GB-2** PIECES. PLACE THE NEXT TWO EQUIDISTANT FROM MIDDLE GB-2 PIECE TO END GB-2 PIECE.

4. SECURE THE **(2)** GB-2 PIECES USING AN IMPACT DRIVER WITH **(8)** 2.5" SELF DRIVING EXTERIOR WOOD SCREWS.



4 STEP

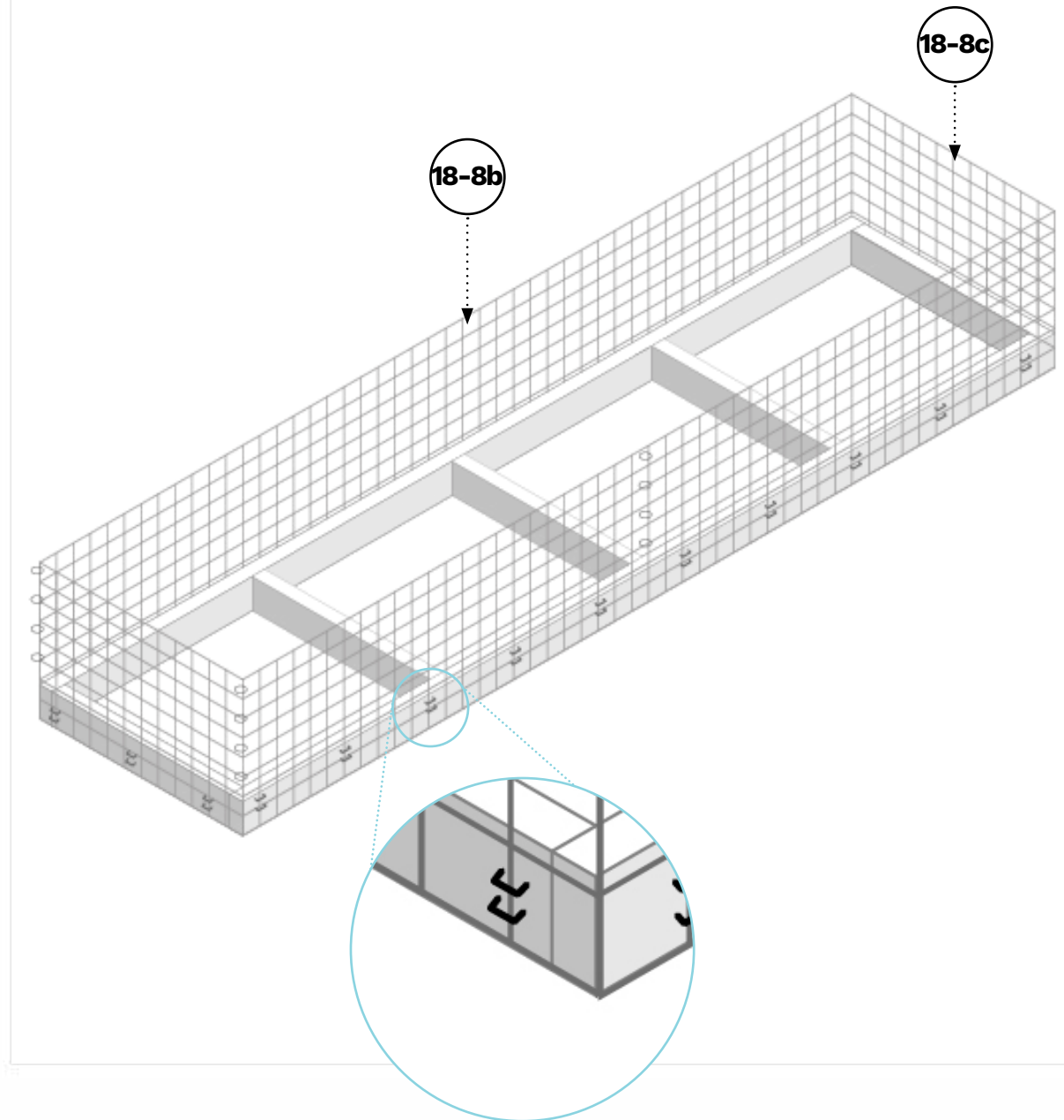
1. SECURE (2) 18-8c AND (2) 18-8b WITH 2X2 WIRE MESH AROUND THE GB-2 AND GB-1 FRAME CREATED IN STEP THREE BY USING (32) 1.5" GALVANIZED STAPLES WITH HAMMER. USE 2 STAPLES PER PIECE OF WIRE, AS SHOWN.

NOTE: SECURE THE 2X2 WIRE MESH ONE SIDE AT A TIME

2. SECURE 2X2 WIRE SHEET USING (2) 18-8c AND (2) 18-8b PIECES TOGETHER BY USING HOG RINGS TO SECURE BASKET TO ITSELF, ONE HOG RING EVERY OTHER OPENING. ENSURE HOG RINGS ARE AT ALL CORNERS AND EDGES



1.5" CABLE STAPLE



5 STEP

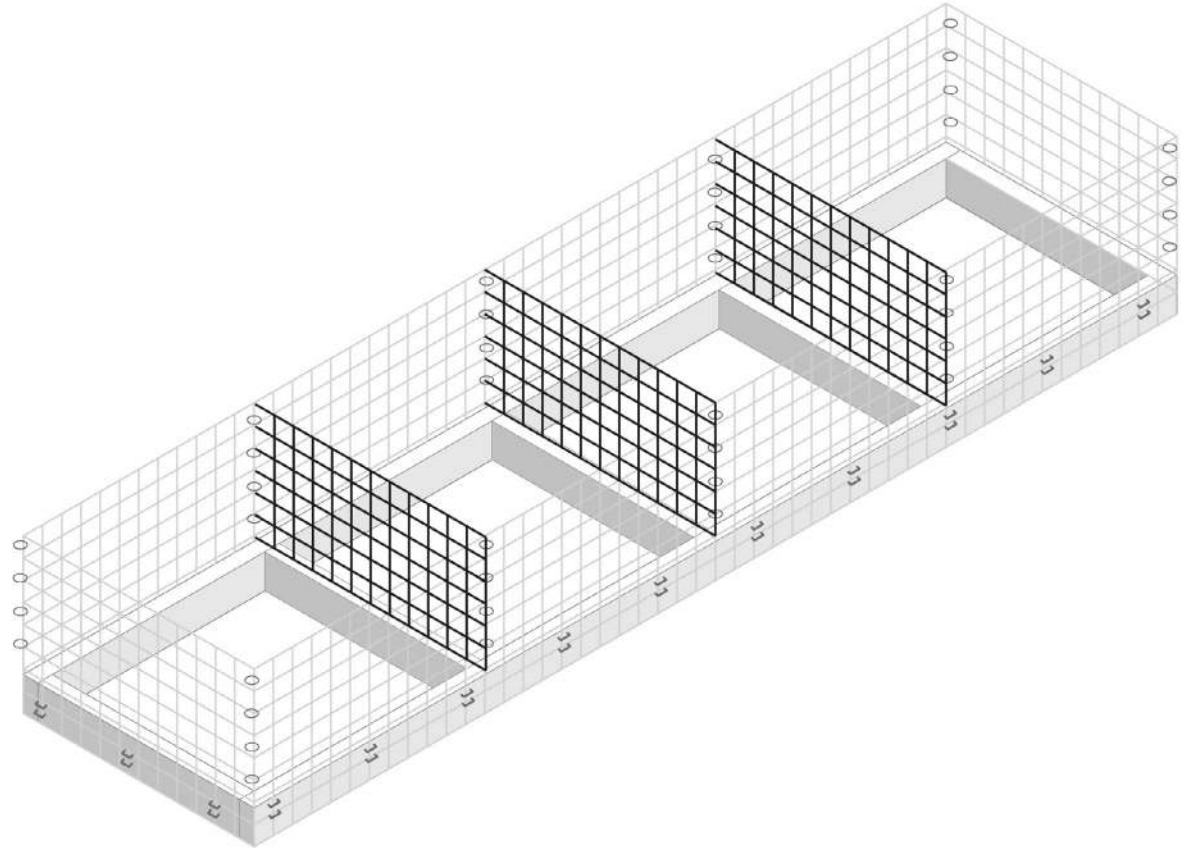
1. USE BRACING MATERIAL

2. IN THE CENTER OF THE GABION BASKET, BELOW THE MIDDLE GB-2 PIECE, USE ONE OF THE FOLLOWING OPTIONS:

OPT A: TAKE EXCESS END PIECES OF 2X2 WIRE METAL MESH + CUT IN HALF WITH BOLT CUTTERS

OPT B: USE ANY EXCESS 2X2 METAL MESH THAT SPANS THE ENTIRE WIDTH OF THE BASKET, SO LONG AS IT HAS A MINIMUM OF 2 BOXES IN HEIGHT (4"+ INCH MINIMUM)

3. SECURE MIDDLE SUPPORT PIECE BY USING HOG RINGS

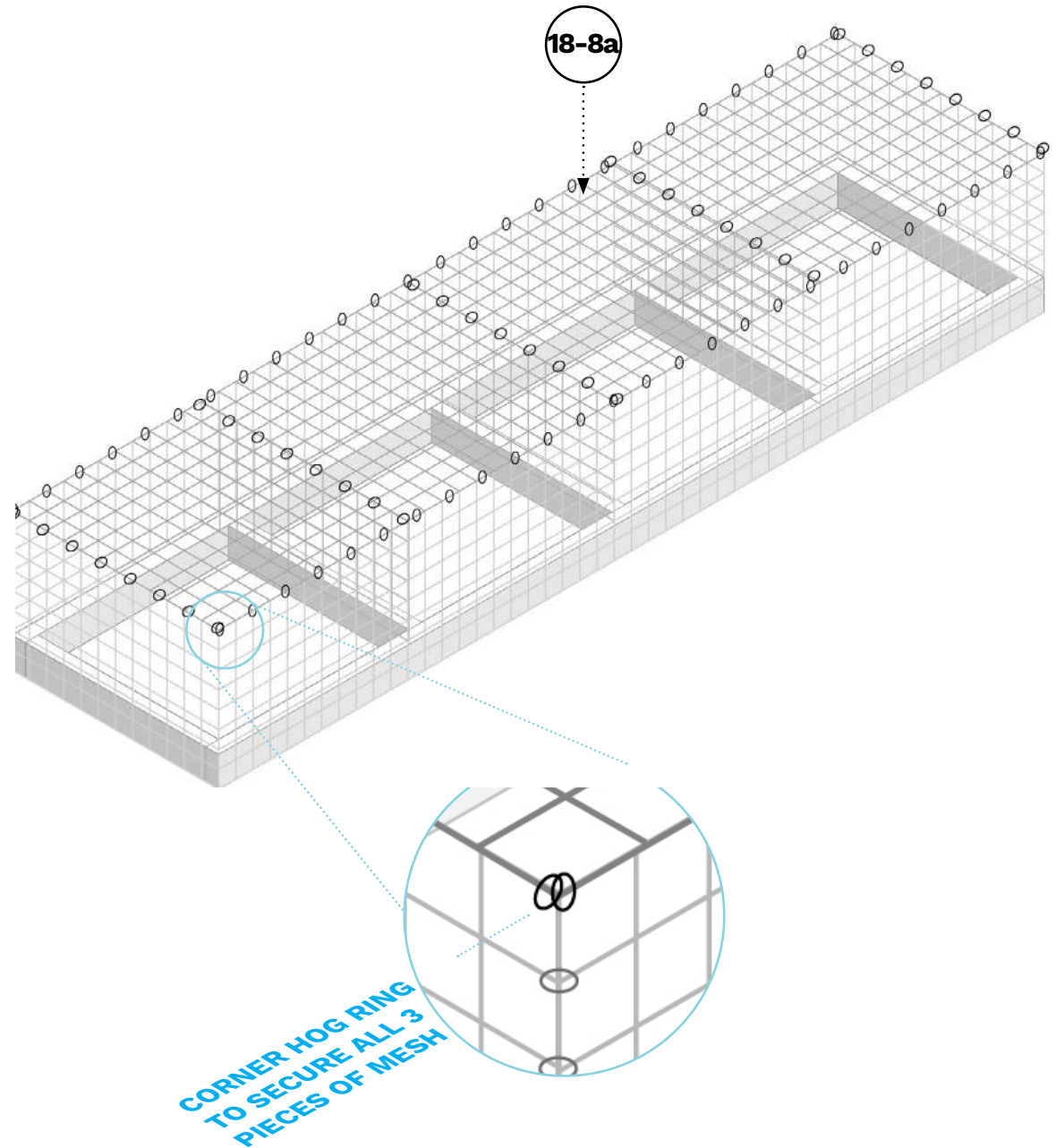


6 STEP

1. PLACE (1) BM - 3 2X2 WIRE MESH TOP PIECE ON TOP OF THE GABION BASKET SECURED IN STEP FIVE.

2. SECURE 2X2 WIRE MESH TOP PIECE BY USING THE HOG RINGS ON THE FOUR SIDES AND THE CENTER BRACE. SECURE 1 HOG RING EVERY OTHER MESH OPENING AND (2) AT EACH CORNER

NOTE: GABION BASKET IS NOW READY TO TRANSPORT TO SITE



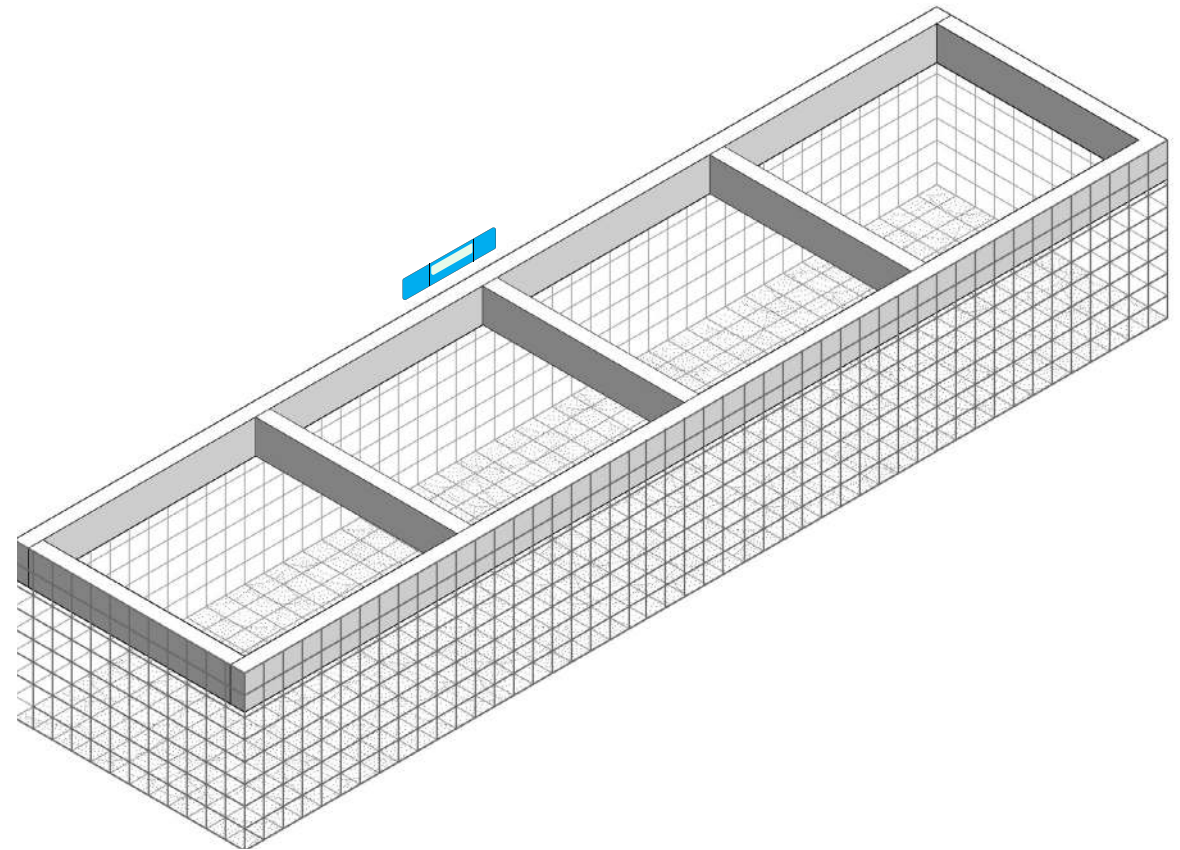
7 STEP

1. ROTATE GABION BASKET 180°.
2. PLACE ON-SITE WITH GRAVEL UNDERLAY. SEE FIG 1
3. ENSURE GABION BASKET IS STRAIGHT USING A LEVEL.

NOTE: IF NEXT TO OTHER GABIONS, USE STRING AND LINE LEVELS TO ENSURE GABIONS ARE STRAIGHT



FIG 1
LEVEL GABION BASKET
WITH GRAVEL



8 STEP

1. PLACE GABION IN ITS LOCATION PER YOUR PARKS DESIGN. APPLY GRAVEL IN BOTTOM TO FINE TUNE ADJUSTMENTS, SEE FIG 1 IN STEP 7.

NOTE: REFERENCE PHASE FOUR IN YOUR CREATE WORKBOOK.

2. STARTING WITH THE LARGEST, FLATTEST PICES OF GABION FILL YOU HAVE, BEGIN FILLING GABIONS, STARTING AT THE EDGES. CONTINUE UNTIL YOU HAVE BUILT UP THE SIDES, SEE FIG. 2

3. FILL THE FRAME WITH LEFTOVER **GABION FILL** UNTIL IT IS LEVEL WITH THE TOP OF THE WOOD EDGE.

NOTE: CAREFULLY PLACE FILL INSIDE BASKET TO AVOID WARPING AND ENSURING ALL GAPS ARE FILLED FOR STABILITY

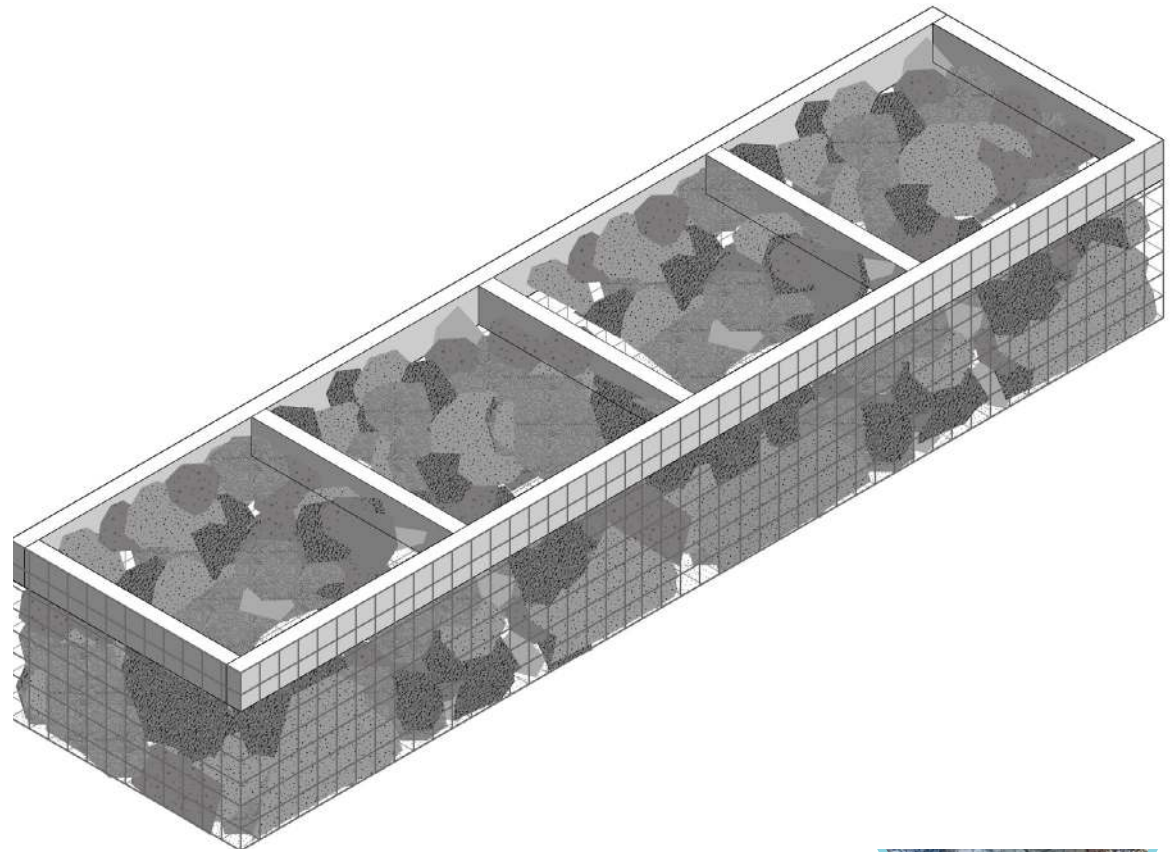


FIG 2

GABION BASKET FILLING ALONG EDGES



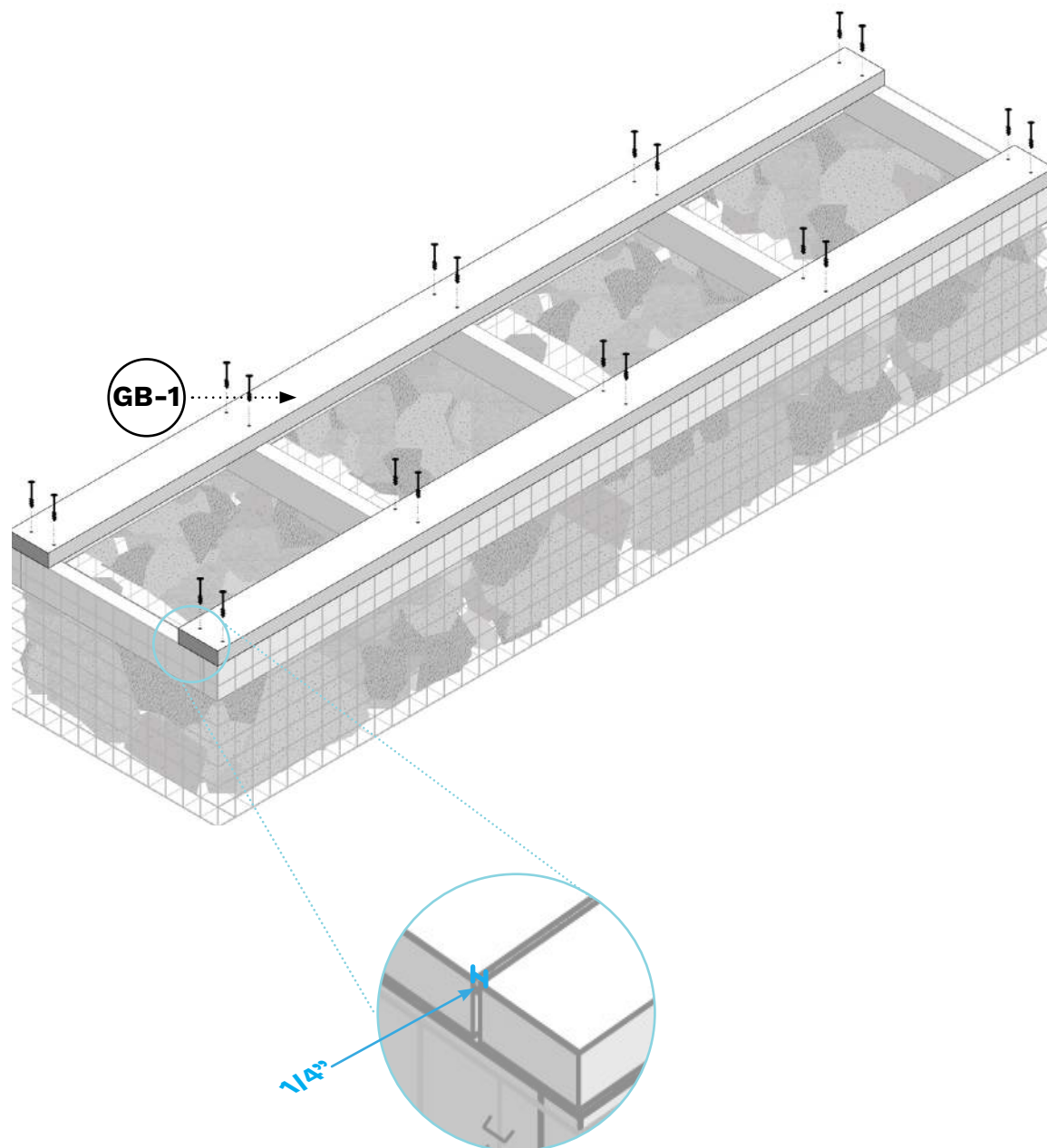
10 STEP

1. PLACE (2) GB-2 PIECES ON TOP OF THE FRAME CREATED IN STEP SIX, USE THE PRE-DRILLING GUIDE, FOUND AT THE END OF THIS DOCUMENT, TO PRE-DRILL THE WOOD. SECURE TWO ENDS FIRST WITH (12) 2.5" SELF DRIVING EXTERIOR WOOD SCREWS, MAKING SURE TO ALIGN EDGE FLUSH WITH WOOD BELOW.

NOTE: PREDRILLING AT THIS STEP IS HELPFUL TO ENSURE SCREWS DON'T SPLIT THE BOARDS.

TIP: ALL SCREWS SHOULD BE A THUMB'S WIDTH AWAY FROM THE EDGE OF THE WOOD, USING THE PRE-DRILLING GUIDE CAN HELP HERE

3. LIGHTLY SAND THE ROUGH EDGES OF THE GABION BENCH. ENSURE NO SHARP EDGES OF GABION WIRE, BUFF WITH GRINDER IF ANY ARE TOO



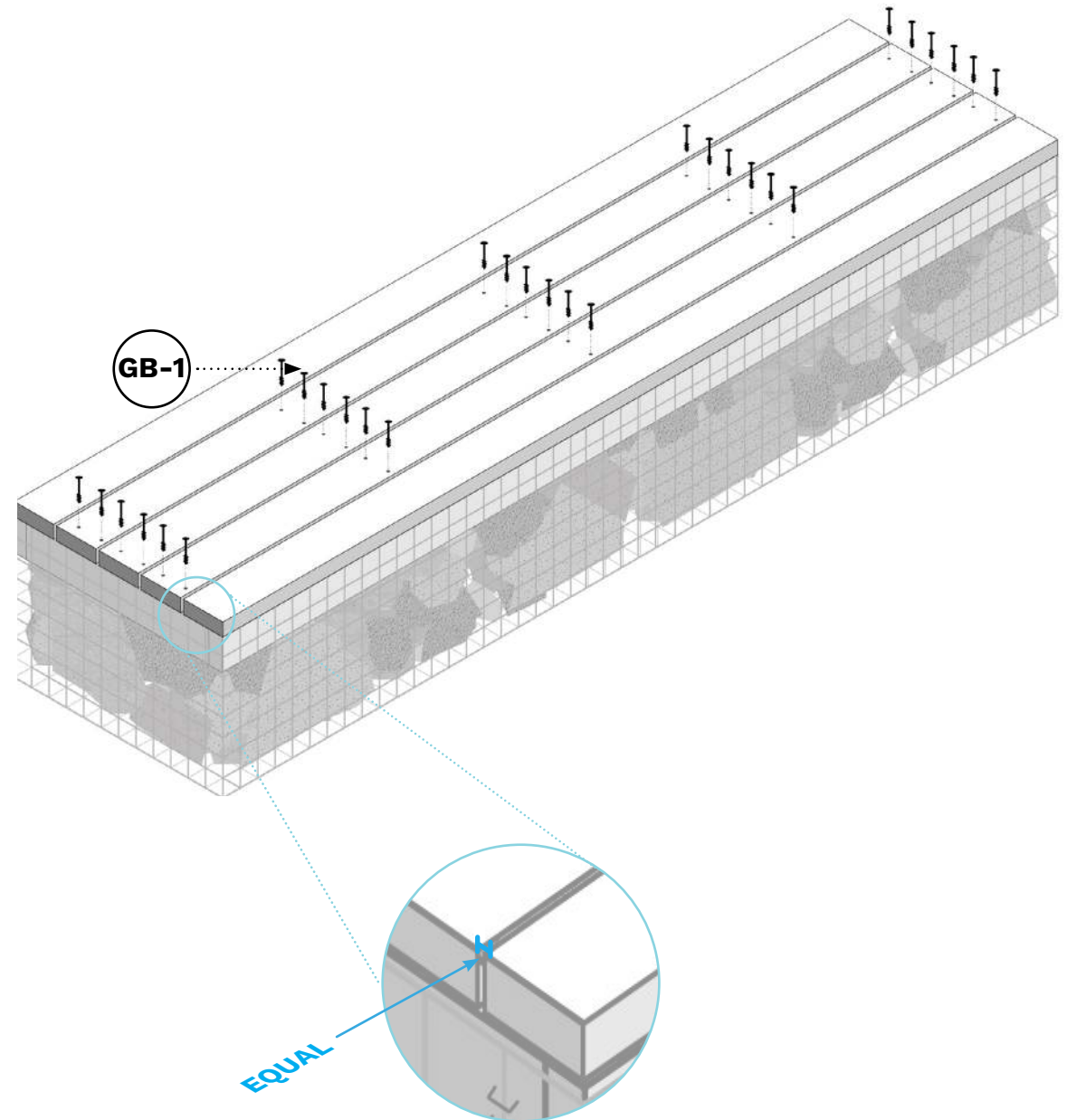
11 STEP

1. EQUALLY SPACE THE OTHER 3 BOARDS BETWEEN TWO END BOARDS. SECURE THE **GB-2 LUMBER USING AN IMPACT DRIVER WITH **(20)** 2.5" SELF DRIVING EXTERIOR WOOD SCREWS.**

NOTE: *PREDRILLING AT THIS STEP IS HELPFUL TO ENSURE SCREWS DON'T SPLIT THE BOARDS.*

TIP: *ALL SCREWS SHOULD BE A THUMB'S WIDTH AWAY FROM THE EDGE OF THE WOOD, USING THE PRE-DRILLING GUIDE CAN HELP HERE*

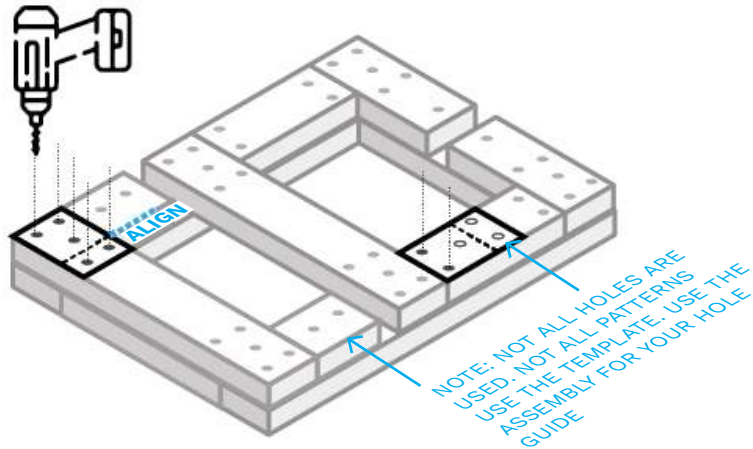
3. LIGHTLY SAND THE ROUGH EDGES OF THE GABION BENCH. ENSURE NO SHARP EDGES OF GABION WIRE, BUFF WITH GRINDER IF ANY ARE TOO SHARP..



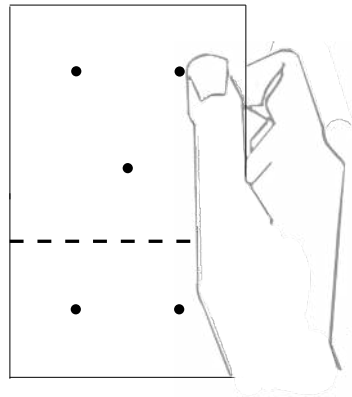


1. CUT OUT PRE-DRILLING HOLE GUIDE ALONG DOTTED LINES

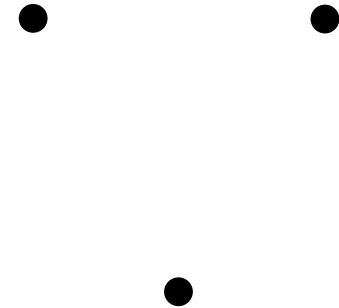
2. WHEN DRILLING, ENSURE THAT YOU PROPERLY **ALIGN THE EDGE OF THE TEMPLATE WITH OUTSIDE EDGE OF THE LUMBER** BEFORE PRE-DRILLING



3. ENSURE ALL HOLES ARE ABOUT A **THUMB IN WIDTH FROM THE EDGE (~1")** EVEN HOLES THAT DON'T USE THIS TEMPLATE



PRE-DRILLING HOLE GUIDE



ALIGN DASHED LINE WITH INSIDE EDGE OF LUMBER



CUT ALONG THE DOTTED LINES

